Comparing counts: Chi-Squared

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Summary

So far we have techniques to study relationships between categorical vrs. numerical variables (t-test, ANOVA).

We are going to study a technique to analyze categorical vrs. categorical relationships.

It's called **chi-squared** (χ^2).

Categorical versus categorical variables

Let's stay we are examining the Latin versions of the gospels of Matthew and John. We are looking for how the 3rd person of "to say" is represented:

Present tense: *dicit* "(he/she) says"
Perfect tense: *dixit* "(he/she) said"

Categorical versus categorical variables

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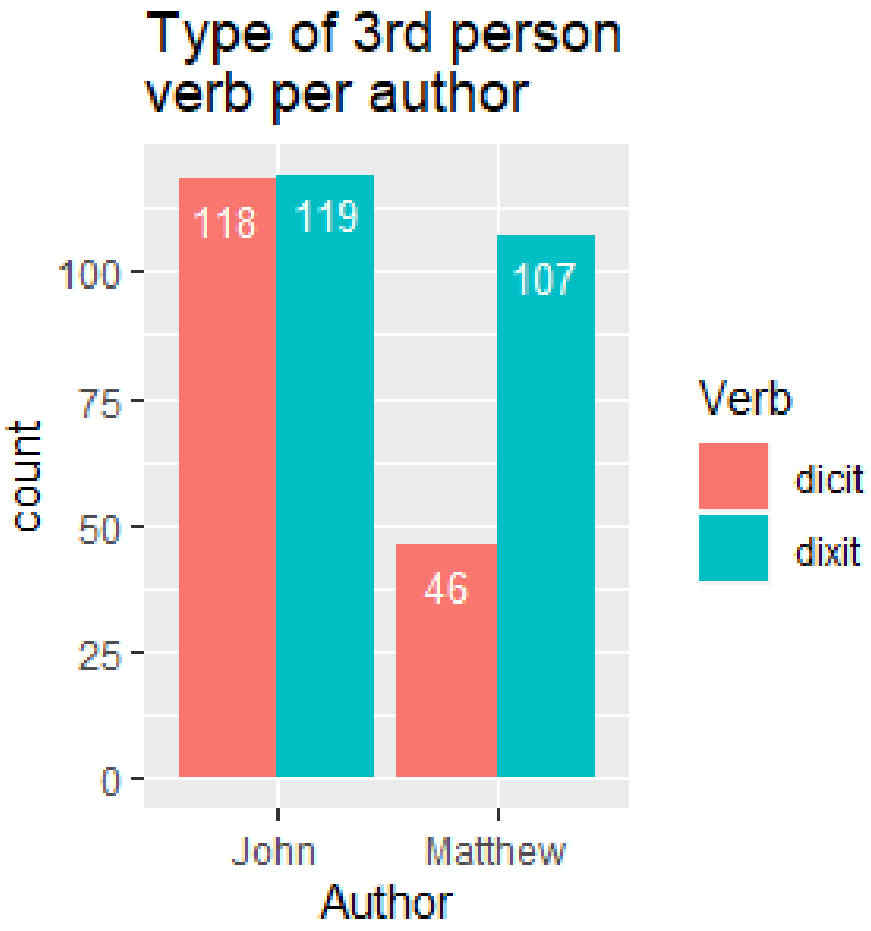
Perfect tense: *dixit* "(he/she) said"

Variable1: Author {Matthew, John}
Variable2: Verb {dicit, dixit}

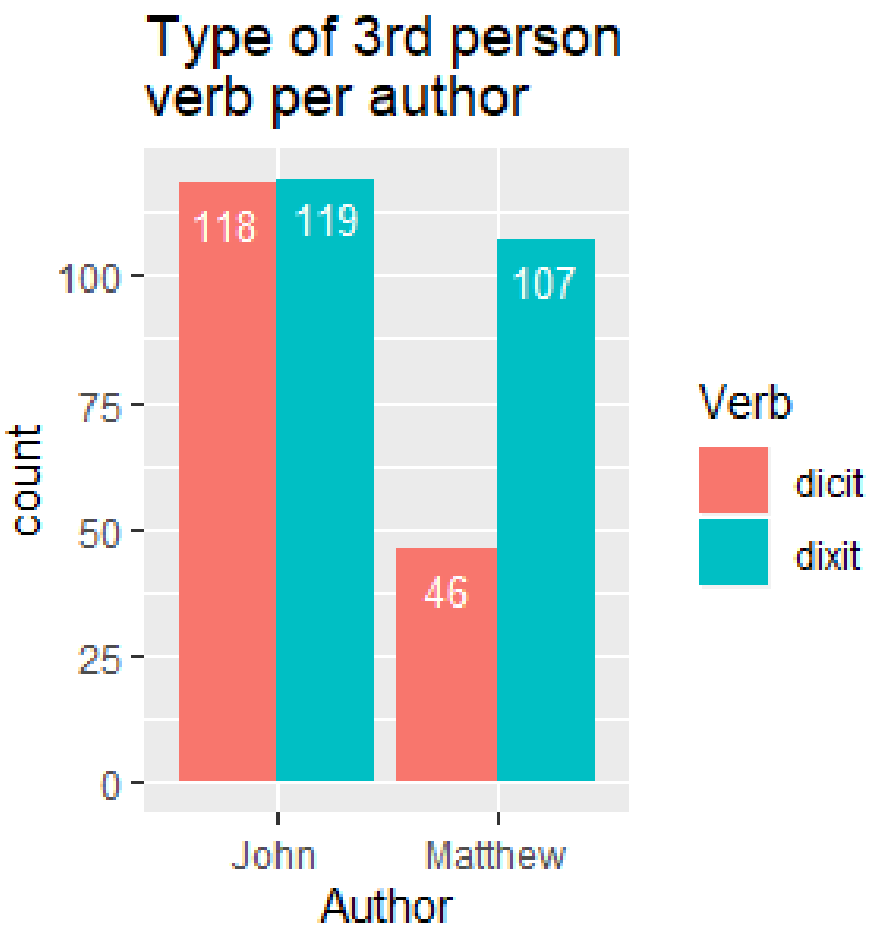


Categorical versus categorical variables

Let's stay we are examining the Latin versions of the gospels of Matthew and John. We are looking for how the 3rd person of "to say" is represented:

A table of frequencies (with two categorical variables) is called a **contingency table**.

Gospel	# occurences *dicit*	# occurences *dixit*
Matthew	46	107
John	118	119

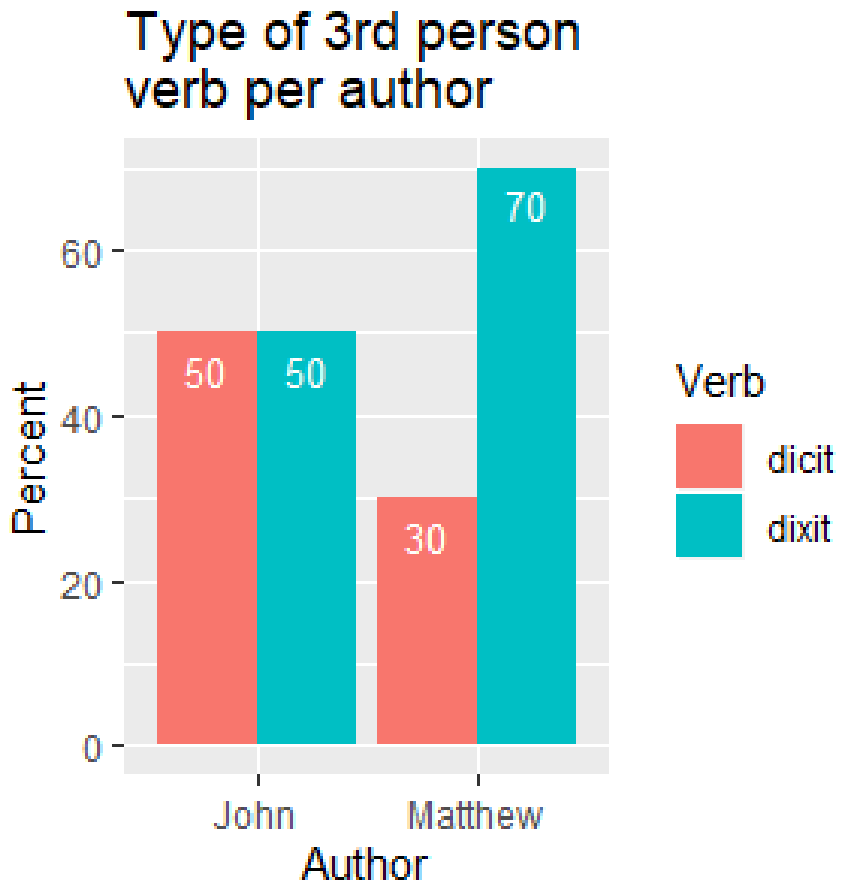


Categorical versus categorical variables

Let's stay we are examining the Latin versions of the gospels of Matthew and John. We are looking for how the 3rd person of "to say" is represented:

A table with percentages is a **proportion table**. We have already used those in R.

Gospel	# occurences *dicit*	# occurences *dixit*
Matthew	46 (30%)	107 (70%)
John	118 (50%)	119 (50%)



Categorical versus categorical variables

R has a test to calculate the differences between these cells. It's called chi-square.

```
> chisq.test(matrixLatin)

	Pearson's Chi-squared test with Yates' continuity
correction

data:  matrixLatin
X-squared = 14.045, df = 1, p-value = 0.0001785
```

If this is less than 0.05, then the two variables are significantly associated.

Categorical versus categorical variables

How does this number work? It calculates the
"expected" versus "observed" frequencies in the data.

Document	# occurences dicit	# occurences dixit	Row totals
Matthew	Obs_{11}=46	Obs_{12}=107	153
John	Obs_{21}=118	Obs_{22}=119	237
Column totals	164	226	390

Categorical versus categorical variables

How does this number work? It calculates the "expected" versus "observed" frequencies in the data.

Document	# occurences dicit		# occurences dixit		Row totals
Matthew	Obs_{11}=46	$E_{11} = \frac{153*164}{390} = 64.3$	Obs_{12}=107	$E_{12} = \frac{153*226}{390} = 88.7$	153
John	Obs_{21}=118	$E_{21} = \frac{237*164}{390} = 99.7$	Obs_{22}=119	$E_{22} = \frac{237*226}{390} = 137.3$	237
Column totals	164		226		390

$$\mathrm{E_{ij}} = \frac{\mathrm{rowTotal} * \mathrm{columnTotal}}{\mathrm{grandTotal}}$$

Categorical versus categorical variables

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Matthew	Obs_{11}=46	Exp_{11}=64	Obs_{12}=107	Exp_{12}=89	153
John	Obs_{21}=118	Exp_{21}=100	Obs_{22}=119	Exp_{22}=137	237
Column totals	164		226		390

Exp: 100 / 64 = 1.55

Exp: 137 / 89 = 1.55

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Exp: 100 / 64 = 1.55
Obs: 118 / 46 = 1.56

Exp: 137 / 89 = 1.55
Obs: 119/107 = 1.11

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$$X^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}.$$

Categorical versus categorical variables

R has a test to calculate the differences between these cells. It's called chi-square.

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```

df = (rows-1)(columns-1) = 2-1 * 2-1 = 1

Categorical versus categorical variables

```
> latin
   Author  Verb count
1 Matthew dicit    46
2 Matthew dixit   107
3    John dicit   118
4    John dixit   119

> matrixLatin = xtabs(count ~ Author+Verb, data=latin)

> matrixLatin
         Verb
Author    dicit dixit
  John      118   119
  Matthew    46   107
```

Categorical versus categorical variables

```
> prop.table(matrixLatin,1)
        Verb
Author        dicit     dixit
  John    0.4978903 0.5021097
  Matthew 0.3006536 0.6993464

> assocstats(matrixLatin)
                    X^2 df   P(> X^2)
Likelihood Ratio 15.115  1 0.00010114
Pearson          14.843  1 0.00011683

Phi-Coefficient   : 0.195
Contingency Coeff.: 0.191
Cramer's V        : 0.195
```

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Phi-Coefficient   : 0.195
Contingency Coeff.: 0.191
Cramer's V        : 0.195
```

The ϕ-coefficient is a measure of how closely these two variables are associated.

Categorical versus categorical variables

Very high ϕ
ϕ = 1
High association

Document	# occurences *dicit*	# occurences *dixit*
Matthew	100	0
John	0	100

Very low ϕ
ϕ = 0
No association

Document	# occurences *dicit*	# occurences *dixit*
Matthew	50	50
John	50	50

How to report a chi-square

Document	# occurences dicit	# occurences dixit
Matthew	46 (30%)	107 (70%)
John	118 (50%)	119 (50%)

```
> chisq.test(matrixLatin)

        Pearson's Chi-squared test with Yates' continuity correction

data:  matrixGospel
X-squared = 14.045, df = 1, p-value = 0.0001785
```

There is a significant association between the documents and their use of tense in Latin verbs ($\chi^2(1) = 14.0$, $p<0.005$, $\phi=0.19$): In the Latin translation of John's gospel, 50% of the verbs "to say" are dicit, while 50% are dixit. On the other hand, in the translation of Matthew's gospel, 30% of the verbs "to say" are dicit, while 70% are dixit.

How to report a chi-square

Document	# occurences dicit	# occurences dixit
Matthew	46 (30%)	107 (70%)
John	118 (50%)	119 (50%)

```
> chisq.test(matrixLatin)

        Pearson's Chi-squared test with Yates' continuity correction

data:  matrixGospel
X-squared = 14.045, df = 1, p-value = 0.0001785
```

There is a significant association between the translator of the gospel and the use of tense in Latin verbs ($\chi^2(1) = 14.0$, $p<0.005$, $\phi=0.19$): John's translation uses approximately the same number of dicit and dixit, whereas Matthew uses 2.3 dixit for every 1 dicit.

Chi-Squared Assumptions

- The sample is randomly selected.
- The observations are independent from one another.
- Every observation belongs to a single category.
- You must have at least 5 elements in each of the “expected” elements table.

Chi-Squared Assumptions

You must have at least 5 elements in each of the “expected” elements table.

```
> chisq.test(matrixLatin)$expected
        Verb
Author       dicit     dixit
  John    99.66154 137.33846
  Matthew 64.33846  88.66154
```

A word of warning

Count data can come in two formats. R handles these types differently.

<u>With the counts already made:</u>

latin

Filter

	Author	Verb	count	perc
1	Matthew	dicit	46	30
2	Matthew	dixit	107	70
3	John	dicit	118	50
4	John	dixit	119	50

<u>With each individual case separately:</u>

columbus

Filter

	onset	strnumbers	emphatic	gender
1	shtr	1	less	m
2	str	0	less	m
3	shtr	1	less	m
4	str	0	less	m
5	str	0	less	m

A word of warning

<u>With the counts already made:</u>

latin

Filter

	Author	Verb	count	perc
1	Matthew	dicit	46	30
2	Matthew	dixit	107	70
3	John	dicit	118	50
4	John	dixit	119	50

```
> matrixLatin
          Verb
Author     dicit dixit
  John       118   119
  Matthew     46   107
```

```
matrixLatin = xtabs(count ~ Author+Verb, data=latin)
chisq.test(matrixLatin)
```

A word of warning

<u>With each individual case separately :</u>

columbus

Filter

	onset	strnumbers	emphatic	gender
1	shtr	1	less	m
2	str	0	less	m
3	shtr	1	less	m
4	str	0	less	m
5	str	0	less	m

```
chisq.test(columbus$onset, columbus$emphatic)
```

Exercise

ʃtr in Ohio:

Data from 120 native-English speakers in Columbus, OH. It was elicited through the *Rapid Anomalous Survey Technique*. Researchers elicited the data by asking for directions to a known street. Their spontaneous pronunciation of [str]eet and [str]aight is the data without emphasis.

After getting directions to the street, the researchers asked “do you mean X street?”. People’s replies had emphasis on words like [str]eet and [str]aight. This is the data with emphasis. (This study is similar to the *fou[r]th floo[r]* studies in NYC).

Data is from: Durian, David. 2007b. *Getting [S]tronger Every Day?: Urbanization and the socio-geographic diffusion of (str) in Columbus, OH*. NWAV 35 Conference Proceedings. University of Pennsylvania Working Papers in Linguistics 13.2: 65-79.

Fisher's Exact Test

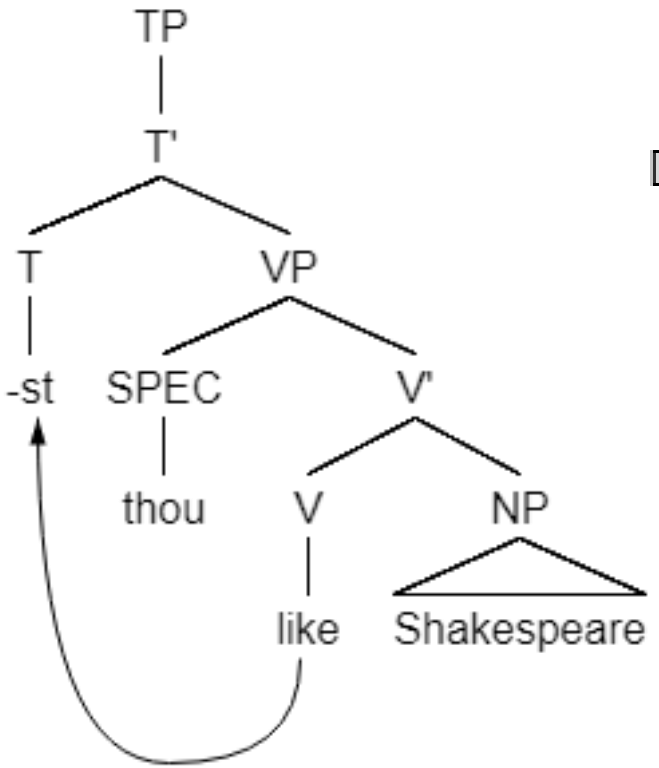
We should only use chi-square **IF the expected values are greater than 5**.

If this assumption is not met, we can use **Fisher's Exact Test**.

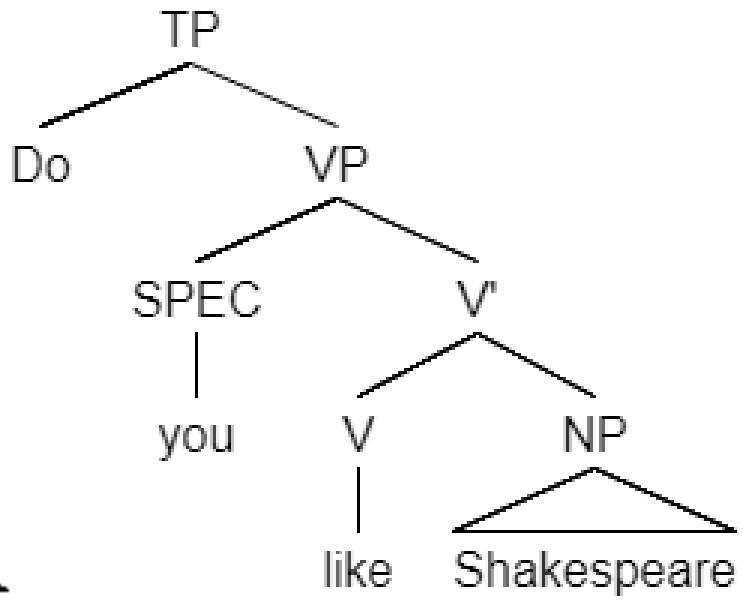
Fisher's Exact Test

	Verb Raising	Do Support
Character 1	1	10
Character 2	3	7
Character 3	2	6
Character 4	4	1

Verb raising (older form)



Do-Support (contemporary form)



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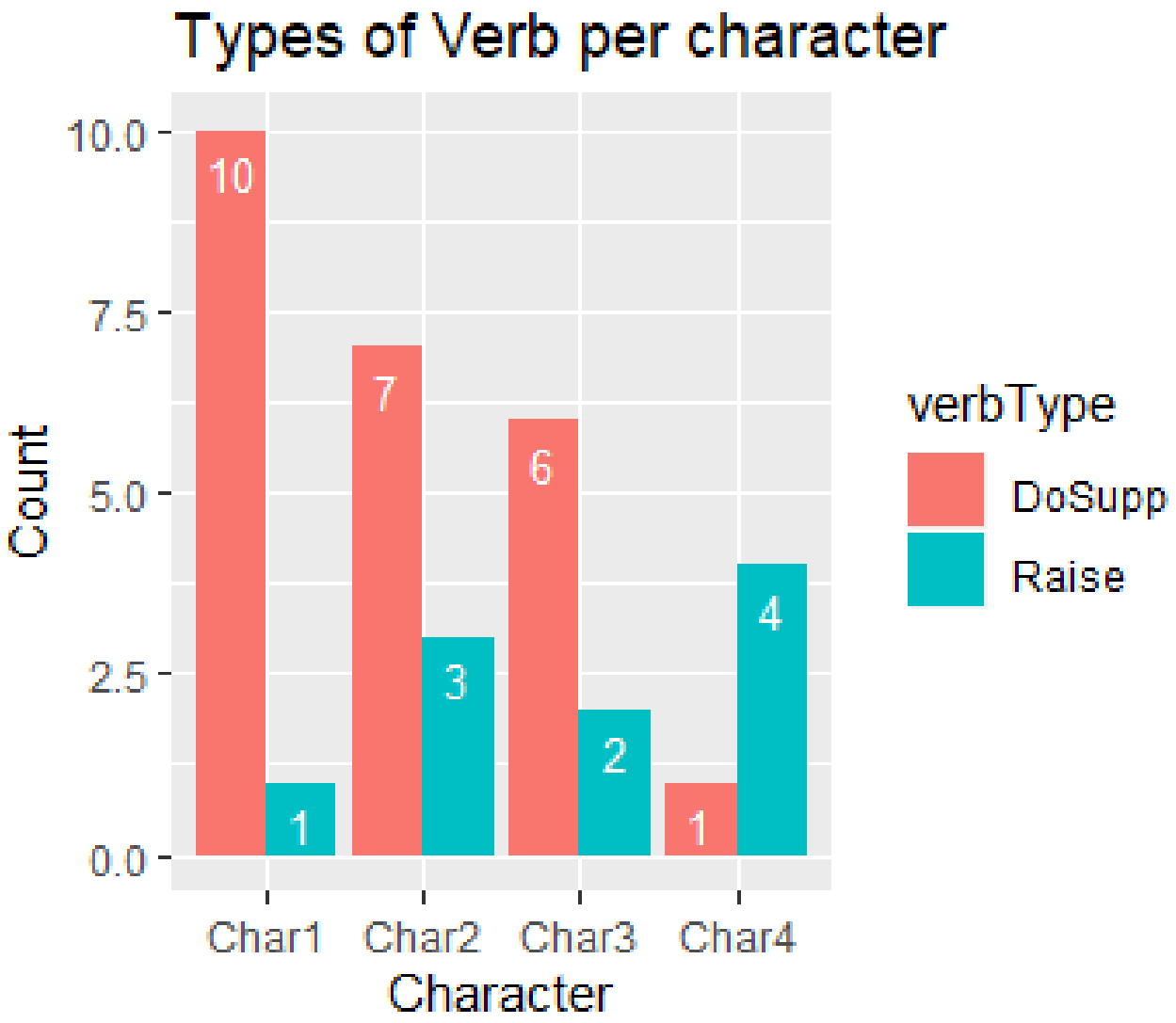

Fisher's Exact Test

	Verb Raising	Do Support
Character 1	1	10
Character 2	3	7
Character 3	2	6
Character 4	4	1

Some of the cells have less than 5 elements. This is a major red flag.

We can get the expected elements like this:

```
> chisq.test(t)$expected
          verbType
character   DoSupp    Raise
    Char1 7.764706 3.235294
    Char2 7.058824 2.941176
    Char3 5.647059 2.352941
    Char4 3.529412 1.470588
```

Fisher's Exact Test

```
> fisher.test(t)

        Fisher's Exact Test for Count Data

data:  t
p-value = 0.04782
alternative hypothesis: two.sided
```

There are significant differences in verb types across characters ($p<0.05$, two-tailed Fisher's exact test). Characters 1, 2 and 3 use more do support, while character 4 uses more verb-raising forms.

Fisher’s Exact Test

When you have so few numbers, it’s more honest to describe the data than to use measurements like effect sizes.

Exercise

Lady Tasting Tea

Summary

Chi-Squared allows you to study the relationship between two categorical variables.

This is particularly useful in areas like sociolinguistics.

We will come back to this idea later when we study logistic regression.